

Elias Wang

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Research Interests

Secure LLM agents under untrusted context: keeping an agent's behavior safe when its inputs — tool outputs, retrieved content, web pages — may be adversarial. I work on how to detect an agent's true intent for security — recovering it across a trajectory of individually benign steps rather than from any single request in isolation.

Education

B.Eng. in Information Security

Sept 2022 – Jul 2026

Northeastern University, Software College — Shenyang, China

- **Selected coursework (grade / 100):** Security Programming (91), Basic Programming Training (90), Object-Oriented Programming & Design (87), Introduction to AI & Big Data (83), Information Content Security (83).

Research

Why Localization Matters: Allocation, Not Detection, for Sparse-Contamination Multi-Source RAG Defense.

Elias Wang (sole author). *Under review* at AISec 2026 (ACM Workshop on Artificial Intelligence and Security, co-located with ACM CCS).

Independent research and undergraduate thesis, 2025–2026.

- Reframed test-time RAG defense under *sparse contamination* (adversarial content confined to a few retrieved segments) as an *allocation* problem, not a detection one; derived a hit-rate decomposition $\mathbb{E}[\text{ASR}] = p_0 - h_{\mathcal{L}}(p_0 - p_1)$ proving the benefit depends only on out-ranking a random allocator — i.e., localization needs a relative ranking, not a calibrated detector.
- Designed a training-free, two-signal localizer (lexical + semantic) driving a localize-then-drop defense that cut average attack success rate from **88.9% to 2.65%** while retaining $\sim 88\%$ of clean-task F1 (0.33 vs. 0.37) at **0.67 $\times$** the undefended token cost — outperforming full-context SmoothLLM on mitigation, utility, and cost *simultaneously*.
- Established attack-generalizability across **5 open-weight LLMs** (Qwen3, Gemma 2, Llama 3, Mistral, Falcon 3) and **2 multi-hop QA benchmarks** (HotpotQA, 2WikiMultiHopQA), against both optimization-based injection (ObliInjection) and fluent knowledge poisoning (CorruptRAG); under poisoning the defense *recovered* clean-task utility the attack had degraded.
- Surfaced and stress-tested an adaptive failure mode (frequency-adaptive payloads), motivating multi-signal localization. Designed, implemented, and evaluated the full pipeline independently.

Experience

LLM Agent Security Development Intern

Mar 2025 – Jul 2025

University–Enterprise Project Training Program, China

- Built a LangGraph-based LLM agent with intent classification, tool routing, and answer generation.
- Designed intent-gated tool routing to enforce least-privilege tool use and reduce unsafe or over-broad agent actions.
- Added safety guards for ambiguous or malformed model outputs: clarification handling, structured action validation, timeout control, and fail-closed fallback behavior.

Honors & Awards

- **First Prize**, NEX Cup Capture-the-Flag (CTF) Competition, Northeastern University.
- **Second-Class Scholarship**, Northeastern University.

Technical Skills

LLM & agents: Python, PyTorch, Hugging Face Transformers, LangGraph, tool-calling / ReAct agent pipelines, RAG & vector search, Model Context Protocol (MCP) servers; Docker, Linux, Git.

Security: agent safety (intent-gating, least-privilege tool allowlists), sandboxed tool execution, prompt-injection & RAG-poisoning threat modeling.

Spoken languages: English (TOEFL iBT 100/120, CEFR C1); Chinese (native).